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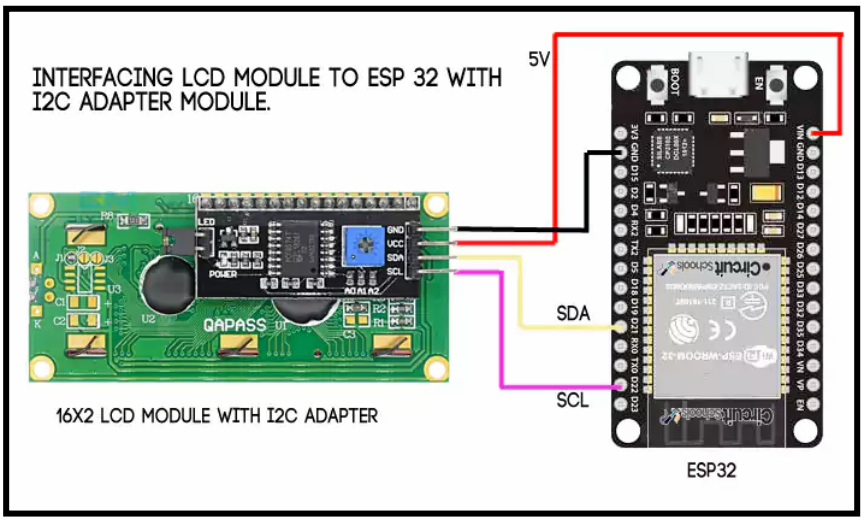
Practical - 05

Aim	Design a sketch to interface LCD with ESP 32 Board.
Software used	Arduino UNO IDE
Hardware used	1. ESP-32
Components required	1. ESP32 2. LCF screen 3. 10K ohm potentiometer 4. 220 ohm resistor 5. Jumper wires 6. Breadboard 7. Pin headers to solder to the LCD Display pin.
Circuit diagram	
Theory	To display text or sensor data using an ESP32, we can connect a **16x2 LCD** (which shows 2 lines of 16 characters each). To make wiring easier and use fewer pins, we use an **I2C module** with the LCD. This way, we only need **two pins** from the ESP32 — **SDA** (data) and **SCL** (clock). The ESP32 sends data to the LCD over the I2C bus using these two pins. We use special libraries in Arduino IDE like Wire.h for I2C communication and LiquidCrystal_I2C.h to control the LCD. The LCD also needs power (VCC and GND) from the ESP32. In the code, we: 1. Set the LCD's I2C address (usually 0x27). 2. Initialize the LCD. 3. Use simple commands like lcd.print() to show text. This setup is commonly used in projects like weather stations, clocks, or any system where we need to show data to the user.

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Code	#include <LiquidCrystal.h>
	int Contrast=75;
	LiquidCrystal led(12,11,5,4,3,2);
	void setup(){
	  analogWrite(6,Contrast);
	  led.begin(16,2);
	
	}
	void loop(){
	  lcd.serCursor(0,0);
	  lcd.print("SHEIKH");
	  lcd.serCursor(0,1);
	  lcd.print("WASIMUDDIN");
	}

Output	
Conclusion	We connected LCD to Arduino Board and displayed our names by verifying and running code in Arduino IDE.